

PRACTITIONERS SURVEY

“Emerging Challenges in Threat Modeling for GenAI-Augmented Systems: A View from the Trenches”

Q.1. Please indicate your gender ['man', 'woman', 'diverse', 'other']
Q.2. Please indicate your age [numeric]
Q.3. Please indicate which of the following roles best describes your work in the organization ['security engineer', 'software engineer', 'data scientist', 'product owner', other (specify)]
Q.4. Please indicate how much experience you have with the system under analysis ['no experience', 'some experience', 'moderate experience', 'extensive experience']
Q.5. Please indicate how extensive your experience is with threat modeling methodologies for security (e.g., STRIDE) ['no experience', 'some experience', 'moderate experience', 'extensive experience']
Q.6. Please indicate your knowledge of security threats affecting GenAI-based software systems ['no prior knowledge', 'minimum knowledge', 'some knowledge', 'moderate knowledge', 'extensive knowledge']
S.1. The method helped identify important security threats for our system
S.2. The threats found by the method were actionable (clear mitigations or next steps)
S.3. The method surfaced threats we would likely have missed otherwise
S.4. The effort required is acceptable for regular use
S.5. The outputs integrate well with our existing tooling (tickets, risk register, architecture docs, CI/CD, security reviews).
S.6. The method adequately captures GenAI-specific risks (e.g., prompt injection, insecure tool use, data leakage)
S.7. The method adequately captures socio-technical risks (e.g., bias, harmful outputs, misuse, human-in-the-loop failure modes)
OE.1. What were the top 3 most valuable threats identified across the methods, and why? [free text]
OE.2. What would have to change for this method to be adoptable in your workflow (training, templates, tooling, ownership, cadence)? [free text]

Note: Statements S.1 to S.7 were asked for all three shortlisted Threat Modeling techniques (three times each). They were assessed through a 6-point Likert values ['completely disagree', 'disagree', 'somewhat disagree', 'somewhat agree', 'agree', 'completely agree']